Question 1

1. a) (i)

$$\overline{\bar{A}+\overline{\left(A\bar{B}\right)}\bar{C}}$$
$$= A\overline{\left(\overline{\left(A\bar{B}\right)}\bar{C}\right)}$$
$$= A\left(A\bar{B}+C\right)$$
$$= A\bar{B}+AC$$

[4]

ii)

$$ABC+B\left(A\oplus B\right)$$
$$= ABC+B\left(\bar{A}B+A\bar{B}\right)$$
$$= ABC+\bar{A}B$$
$$= B\left(AC+\bar{A}\right)$$
$$= B\left(C+\bar{A}\right)$$
$$= BC+\bar{A}B$$

[4]

b)

$$f=\bar{A}\bar{B}\left(\overline{C\oplus D}\right)+A\bar{B}\bar{C}\bar{D}+ACD+ABD$$
$$=\bar{A}\bar{B}CD+\bar{A}\bar{B}\bar{C}\bar{D}+A\bar{B}\bar{C}\bar{D}+ACD+ABD$$

f	CD	00	01	11	10
AB					
00		1	0	1	0
01		0	0	0	0
11		0	1	1	0
10		1	0	1	0

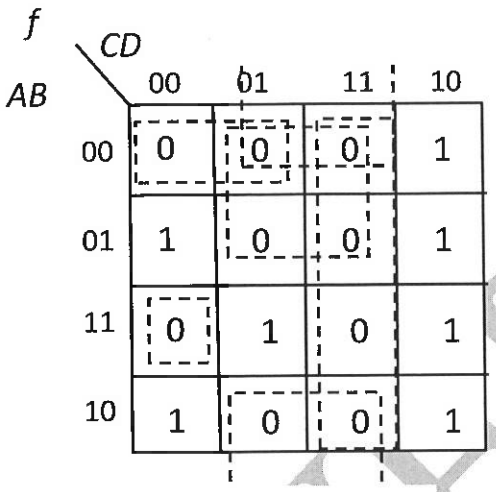
$$f=\bar{B}\bar{C}\bar{D}+ABD+\bar{B}CD$$

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Here, 1 mark for drawing the Karnaugh map, 1 for filling it out correctly, 1 for the correct grouping, and 1 for the final expression.

[4]

c)



$$f = \left(A + \overline{D}\right)\left(B + \overline{D}\right)\left(A + B + C\right)\left(\overline{A} + \overline{B} + C + D\right)\left(B + \overline{C} + \overline{D}\right)$$

Here, 1 mark for drawing the Karnaugh map, 1 for filling it out correctly, 1 for the correct grouping, and 1 for the final expression.

[4]

d)

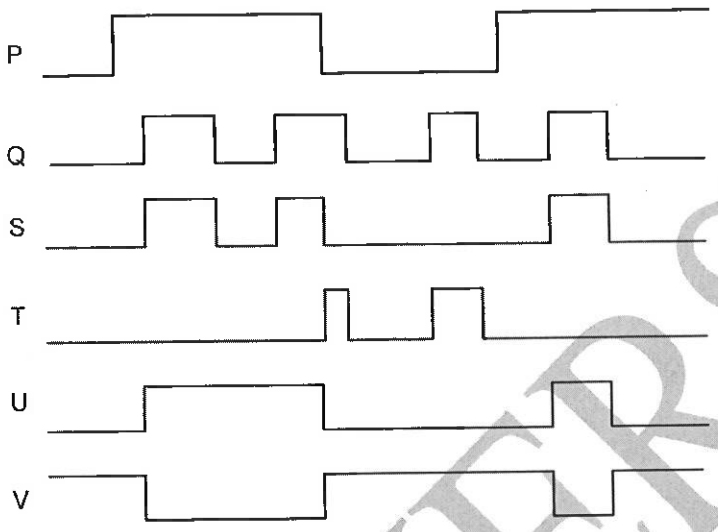
Decimal	Hexadecimal	Binary	BCD
5153	1421		
		0011 0111 1001 1000	(1) 0100 0010 0011 0010
	1CF	0000 0001 1100 1111	
-1101		1111 1011 1011 0011	

Give 2 marks per answer.

[8]

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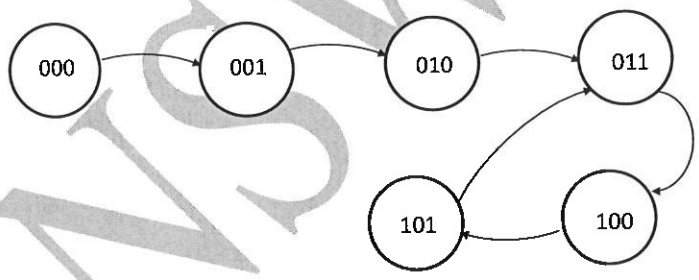
e)



Marks: 1 mark each for S and T and 2 marks each for U and V.

[6]

f)



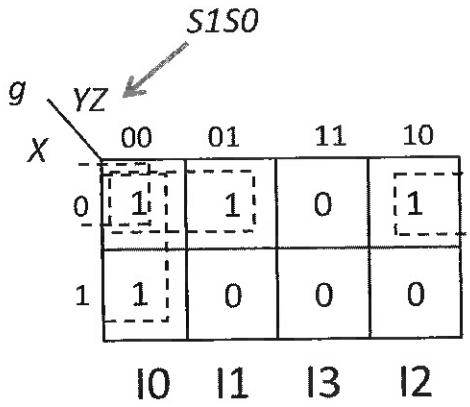
Give 2 marks correct states and 2 marks for correct interconnections.

[4]

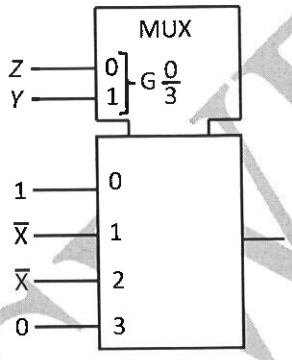
g) In an implementation using a multiplexer, a 4 × 1 multiplexer is sufficient for a three variable logic function.

Assuming variable Y and Z are connected to multiplexer select lines S1 and S0 respectively, the K-map implementation in part (a) allows us to identify:

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Where I[3:0] are the input lines of the multiplexer. This gives us the following solution:



Other combinations of X, Y or Z on S1 and S0 are also possible.
Marks: 2 for selecting correct multiplexer, 2 for logic, and 2 for final circuit.

[6]

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Question 2

(a) (i)

A_0	B_0	C_0	S_0	C_1
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Here 0.5 marks can be awarded per row of the table.

[4]

(ii)

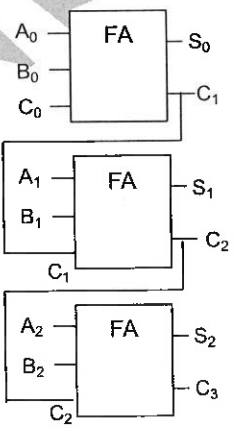
$$S_0 = \overline{A_0}\,\overline{B_0}\,C_0 + \overline{A_0}\,B_0\,\overline{C_0} + A_0\,\overline{B_0}\,\overline{C_0} + A_0B_0C_0 = A_0 \oplus B_0 \oplus C_0$$

$$C_1 = \overline{A_0}B_0C_0 + A_0\overline{B_0}C_0 + A_0B_0\overline{C_0} + A_0B_0C_0 = A_0C_0 + A_0B_0 + B_0C_0$$

2 marks for each expression.

[4]

(iii)



From the 3-bit ripple adder circuit above, the worst-case propagation delay occurs when the final carry C_3 requires the first output carry C_1 to propagate through the circuit (2 marks for seeing this). For the FA implementation in (ii), the propagation delay for both S_0 and C_1 is two gate stages, i.e. 20 ns. Therefore, e.g.

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with input carry $C_0 = 0$, A[2:0] = 111 and B[2:0] = 111, C_3 has a propagation delay of 3 x 20 = 60 ns (2 marks).

[4]

(b) Look ahead carry (LAC) adders are not taught in the course. However, the question provides enough information for the logic for this circuit to allow students to work out the answer.

(i) The first step is to write expressions for G_n and P_n, from the given logic description:

$$G_n = A_n B_n \quad , \quad P_n = A_n + B_n$$

[2 marks]

(ii) Each of these has a propagation delay of 1 gate stage, i.e. 10 ns. To generate the output carry C_{n+1}, either G_n, or P_n and C_n need to be high. It is then possible to write C_{n+1} in terms of G_n, P_n and C_n:

$$C_{n+1} = G_n + P_n C_n$$

[2 marks]

(iii) Using C_{n+1} it is possible to write expressions for C_1, C_2 and C_3, and the propagation delays (two gate stages, 2x10 ns, plus the single gate stage delay of 10 ns for P_n, G_n), as follows:

$$C_1 = G_0 + P_0 C_0$$

Delay: 30 ns

$$C_2 = G_1 + P_1 C_1 = G_1 + P_1 G_0 + P_0 P_1 C_0$$

Delay: 30 ns

$$C_3 = G_2 + P_2 C_2 = G_2 + P_2(G_1 + P_1 G_0 + P_0 P_1 C_0)$$
$$= G_2 + P_2 G_1 + P_1 P_2 G_0 + P_0 P_1 P_2 C_0$$

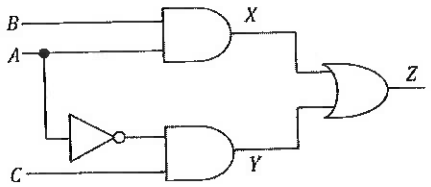
Delay: 30 ns

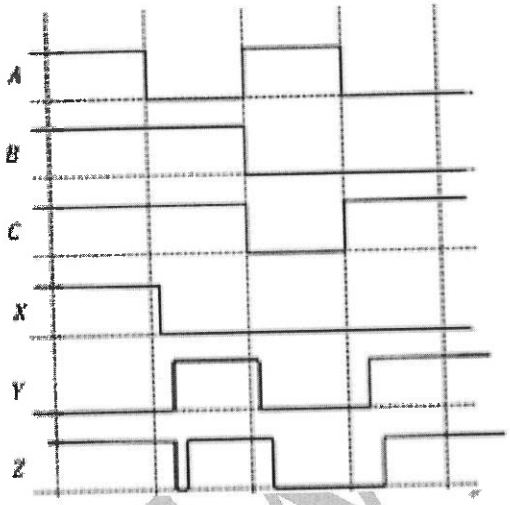
[4 marks]

Note that the propagation delay for C_3 = 30 ns is faster than the 60 ns worst-case result for part (a). Furthermore, the 'sum' terms, S_n take 20 ns (S_0), 50 ns (S_1) and 50 ns (S_2), also less than the worst-case 60 ns result for part (a).

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c) (i)





1 mark each for the waveforms for X and Y and 4 marks for output Z. These should indicate the switching delay, and the glitch in Z.

[6]

(ii) To prevent a static 1 hazard, i.e. a glitch which transitions from 1 – 0 – 1, we require covering '1's in the Karnaugh map which are adjacent but in different groups. For the function in the circuit:

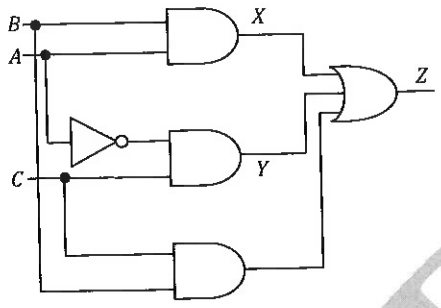
$$Z = AB + \bar{A}C$$

The Karnaugh map is as follows:

Z	BC 00	01	11	10
A 0	0	1	1	0
1	0	0	1	1

Here the grouping BC will eliminate the hazard, giving the Boolean expression and circuit as follows:

$$Z = AB + \bar{A}C + BC$$



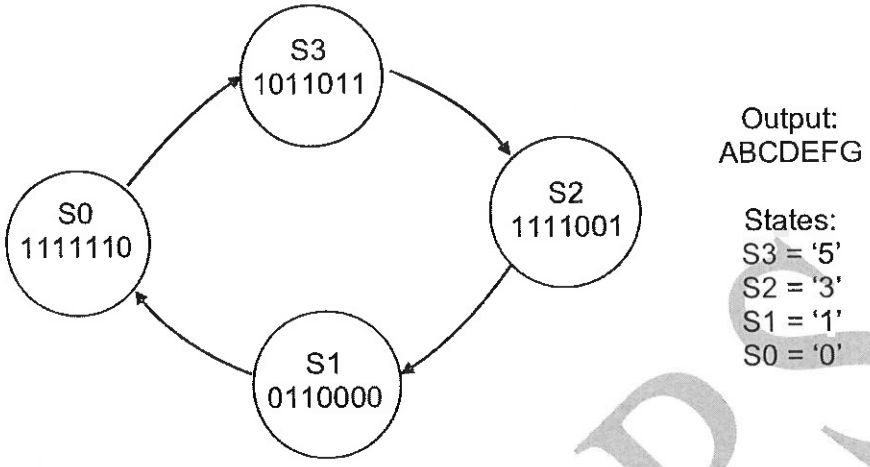
2 marks for Karnaugh map, 1 mark for the introduction of the correct extra term or Boolean expression, and 1 for the redesigned circuit diagram.

[4]

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Question 3

a) Moore state diagram:



Give 2 mark for four states and interconnections, 2 mark for correct labels/values.

[6]

b) State transition table:

Current state value	Current state							Next state						
	A	B	C	D	E	F	G	A^+	B^+	C^+	D^+	E^+	F^+	G^+
5	1	0	1	1	0	1	1	1	1	1	1	0	0	1
3	1	1	1	1	0	0	1	0	1	1	0	0	0	0
1	0	1	1	0	0	0	0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0	1	0	1	1	0	1	1

Give 2 marks per row.

[8]

c) As there are 7 variables, a Karnaugh map approach to obtain simplified Boolean expressions is very difficult. However, direct inspection of the state transition table may be used to identify the following Boolean expressions (3 marks for seeing this approach):

$A^+ = F + \bar{D}$ $\quad B^+ = \bar{E}$ $\quad C^+ = 1$

$D^+ = A^+$ $\quad E^+ = \bar{D}$ $\quad F^+ = \bar{G}$

$G^+ = F$

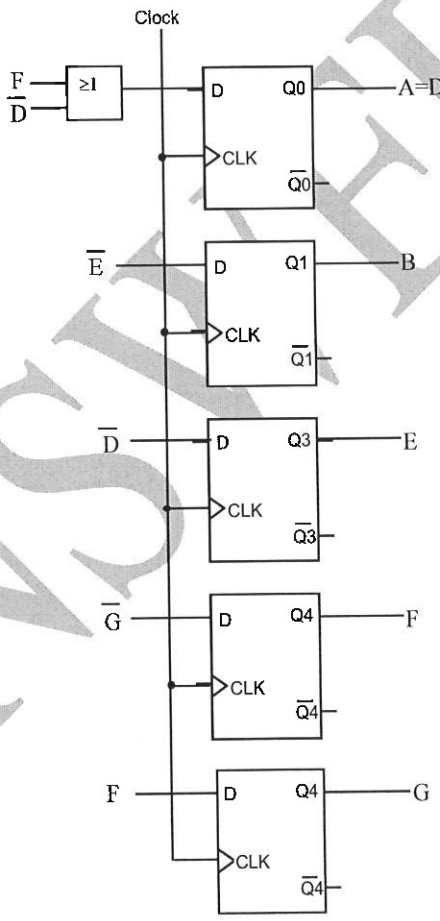
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Give 1 mark per expression. Alternative solutions may also exist, give full marks for minimal solutions.

Students may solve this question with a two-bit state assignment for the four states S0-S3, and subsequent logic/decoding for to drive the 7 segment display. In this case, up to 8 marks may be given.

[10]

d) Independent flip-flops for C and D may be eliminated, as C = 1 and D = A (2 marks for this). We may also use the inverted outputs of flip-flops, where possible, to eliminate any inverters. The circuit then uses 5 flip-flops and one OR gate (4 marks for the circuit). If an alternative state assignment and decoding solution is used, up to 4 marks may be awarded.



[6]

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